Sub-Section Number : 1
Sub-Section Id : 640653214974
Question Shuffling Allowed : No

Question Number : 312 Question Id : 6406531326619 Question Type : MCQ
Correct Marks : 0
Question Label : Multiple Choice Question
THIS IS QUESTION PAPER FOR THE SUBJECT "DEGREE LEVEL : SPEECH TECHNOLOGY (PEN AND PAPER EXAM)"

ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT?
CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN.

(IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)
Options :
6406534441671. ✔ YES
6406534441672. ✖ NO

Sub-Section Number : 2
Sub-Section Id : 640653214975
Question Shuffling Allowed : No

Question Number : 313 Question Id : 6406531326620 Question Type : MCQ
Correct Marks : 25
Question Label : Multiple Choice Question

1. (1 point) In a traditional HMM-GMM based speech recognizer, which part models the **probability of an observed acoustic feature vector given a hidden state**?

   A) The Hidden Markov Model (HMM)

   B) The Lexicon Dictionary

   C) The Gaussian Mixture Model (GMM)

   D) The Language Model (LM)

2. (1 point) In an HMM-GMM based speech recognition system, what is the primary purpose of the **lexicon** (pronunciation dictionary)?

   A) To model the probability distribution of acoustic features.

   B) To specify the acoustic model parameters for each HMM state.

   C) To define how words are broken into sequences of phonemes.

   D) To map acoustic feature vectors directly to word sequences without intermediate units.

3. (1 point) What happens if a neural network is built with only linear layers and no non-linear activation functions?

   A) The network can approximate any complex non-linear function.

   B) The network still learns non-linear decision boundaries automatically.

   C) The network is equivalent to a single linear transformation.

   D) The network performs better because non-linearities add noise.

4. (1 point) Suppose you are training a neural network to classify which **phoneme** a particular audio frame or segment belongs to. You have a dataset of speech where each input frame is labeled with its correct phoneme class.

   Which loss function is most appropriate for this task?

   A) Mean Squared Error (MSE) Loss

   B) Binary Cross-Entropy Loss

   C) Categorical Cross-Entropy (Softmax Cross-Entropy) Loss

   D) Hinge Loss

5. (3 point) A single-layer neural network takes **3 inputs** $x = [x_1, x_2, x_3]$ and produces one output with a **sigmoid activation**.

   The weights and bias are:

$$w = [0.2,\ -0.4,\ 0.1], \quad b = 0.5.$$

   Given:

$$x = [2,\ 1,\ -1].$$

   The output is:

$$y = \sigma(z) = \frac{1}{1+e^{-z}}, \quad z = w \cdot x + b.$$

   (a) (1 point) Compute the value of z.

   (b) (1 point) Compute the output y.

   (c) (1 point) Compute the gradient $\frac{\partial y}{\partial z}$. Show the formula you used.

6. (3 point) Suppose your model outputs the following logits for a 3-class classification task:

$$\mathbf{z} = [2.0,\ 1.0,\ 0.1]$$

   (a) (2 point) Compute the softmax probabilities for each class.

   (b) (1 point) If the true class label is class 1 (index 0), compute the cross-entropy loss.

7. (5 points) Explain all the steps involved in computing Mel-Frequency Cepstral Coefficients (MFCCs).

8. (5 points)

(a) (2 point) Explain the difference between tri-phone and tri-state model of speech recognition.

(b) Suppose you are designing an HMM-GMM based speech recognition system for the word **"cat"**.

- You use **triphone** modeling (context-dependent phones).
- Each phone is modeled as a **3-state HMM** (tri-state).

The phonemes in "cat" are: /k/ – /ae/ – /t/.

(a) (1 point) How many HMM states will there be for the word "cat" if you model each phoneme as a monophone with 3 states?

(b) (1 point) If you instead use **triphones**, how many unique context-dependent phone units does "cat" contain? Write out their left and right contexts. Assume a silence `/sil/` at the start and end.

(c) (1 point) How many total HMM states would this triphone model use for "cat"?

9. (5 point) A neural network predicts the following per-timestep probability distributions over the label set $\{A, B, -\}$, where "–" is the blank label.

You have **3 time steps** with output probabilities:

Timestep	$P(A)$	$P(B)$	$P(-)$
$t = 1$	0.1	0.6	0.3
$t = 2$	0.7	0.1	0.2
$t = 3$	0.1	0.7	0.2

The target transcription is the label sequence $[B, A]$.

(a) (1 point) List all valid alignments that collapse to $[B, A]$ using the CTC collapse rule (remove repeated labels and blanks).

(b) (2 point) Compute the probability of each valid alignment.

(c) (2 point) Compute the total probability for the target transcription and the CTC loss:

$$L = -\log(\text{total probability})$$

Hint: A valid alignment must produce $[B, A]$ after collapsing. For example, both $[B, -, A]$ and $[B, B, A]$ are valid.

Options :

6406534441673. ✔ I have written answers on the answer sheets

6406534441674. ✖ Not applicable